

Certificate of Compliance — Verification Guideline

How every CoC is marked against SANS 10142-1 · (647) FORTRESS – Asset Audits

Reference standard: SANS 10142-1:2021 (Edition 3.1), Section 8 & clause 5.4.2 · Rule set v0.2 · Issued 19 June 2026 · Prepared by Watson Mattheus Consulting Electrical Engineers

Purpose. This document explains, step by step, how each electrical Certificate of Compliance (CoC) in the audit is reviewed and marked. Use it as the standard reference so that **every CoC is judged the same way**, by the same rules, regardless of who performs the review. The same rule set underlies the DB schedules, the per-COC verdicts, and the failure reports.

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1 · The process at a glance

Each site folder of CoC scans is taken through six stages. Stages 1–2 build the **register** (what exists, classified and renamed); stages 3–5 build the **verification** (does each CoC meet SANS 10142-1).

1

Read certificate

Each scan is read to capture the CoC number, Initial/Supplementary tick, and the DB/tenant it covers.

2

Schedule & rename

A DB schedule is built per site and files are renamed `COC#_DB_I/S`. Non-electrical certs are excluded.

3

Read test report

The Section 4 inspection & test page of each CoC is read — the 4 inspection items and 15 test values.

4

Apply rule set

The SANS 10142-1 thresholds are applied to every item to mark PASS / FAIL / cannot-verify.

5

Verdict

An overall verdict is reached per CoC and written to the verification workbook.

Report

A one-page report is produced for every CoC, plus a site findings summary.

2 · Status & verdict colour key

Two sets of labels are used. **Schedule status** answers “is the right certificate on file for this DB?”. **Verification verdict** answers “does the certificate meet the standard?”.

Verification verdict (per CoC)

PASS

Test report present, signed & dated, all inspection items Yes, and every readable test value meets its SANS limit.

FAIL

A clear breach: an inspection item marked “No”, or a legibly-read test value outside its SANS limit.

REVIEW

No hard breach, but key handwritten values are illegible or a test-report page is missing — a person must read the original.

Schedule status (per DB)

Status	Meaning
OK	An Initial CoC is on file (or a Supplementary correctly linked to an Initial that is on file).
FAIL	Supplementary CoC(s) on file but no Initial — the installation has no base certificate.
FLAG	A Supplementary references an Initial CoC number that is not in the supplied set — request it.
MISSING	The DB requires a CoC but none is on file.
REVIEW	A CoC exists but its Initial/Supplementary box was not ticked — type cannot be confirmed.
N/A	Vacant unit / storeroom / item that does not require a CoC.

3 · Stage A — Classify: Initial vs Supplementary

This is done **first** because it sets the filename marker (**I** or **S**) and decides which cross-checks apply. Three signals on the certificate are read:

- **“Type COC” tick** — *Initial* or *Supplementary* (where the form carries this field).
- **Test-report Section 2** — “Existing Certificate? Yes/No”, with “Date issued” and “Previous COC Number”.

- **Installation-type tick** — New / Complete vs Alteration / Extension / Existing / Only part of installation.

Outcome	When	Marker
INITIAL	“Initial” ticked; OR “Existing Certificate?” = No and no Previous CoC No.; OR New / Complete installation.	I
SUPPLEMENTARY	“Supplementary” ticked; OR a Previous CoC No. is given; OR Alteration / Extension / part of an existing installation.	S
UNCLEAR	Signals conflict or are blank — flag for a manual decision; do not guess.	U

Cross-reference rules for a Supplementary. A Supplementary CoC must record the **Previous/Original CoC Number** it supplements. The referenced Initial should exist in the set and pertain to the same installation/DB/address, and the Supplementary’s date must be on or after the Initial’s. A Supplementary that references **no Initial at all** fails the DB; one that references an Initial which simply was not supplied is **flagged** so it can be requested.

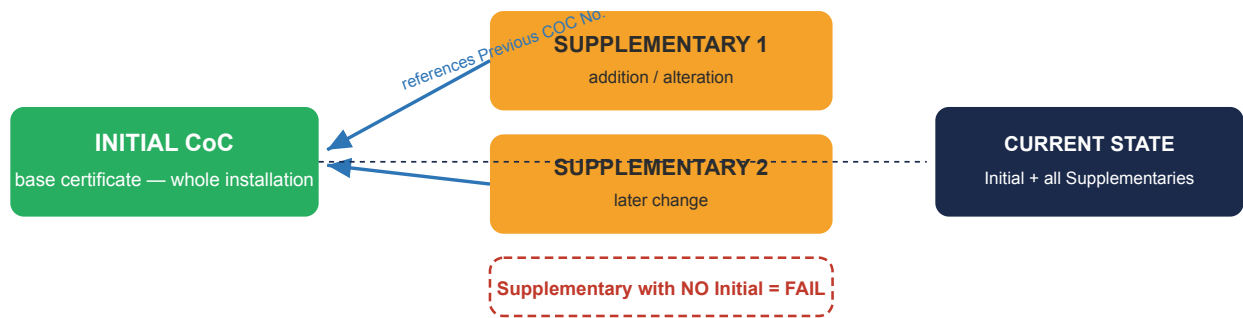
3a · Initial & Supplementary — relationships & procedure

What each one is

Certificate	What it certifies	Issued under
Initial CoC	The base certificate for the whole electrical installation (or the complete DB). It is the foundation record for that installation and remains valid until the installation is materially altered.	EIR 7(1) / OHS Act — new or complete installation [9(2)(a)/(b)]
Supplementary CoC	Issued when an addition or alteration is made to an installation that already holds a CoC. It covers only the new/changed part and must be read together with the Initial — it does not replace it.	EIR 9(2)(c) — new part to an existing installation

The relationship

An installation has **one Initial** as its base. Over time it may gain **several Supplementaries**, each pointing back to that Initial by its *Previous CoC Number*. Together — Initial + every Supplementary — they describe the installation as it stands today. A Supplementary can never stand alone: without an Initial, the base installation is uncertified.



Principles the reviewer applies

- A DB's electrical safety record is only complete with a **valid Initial** as its base.
- Every Supplementary must **reference its Initial** (Previous COC Number + date issued).
- The referenced Initial must be **the same installation/DB/address** as the Supplementary.
- A Supplementary's issue date must be **on or after** the Initial it supplements.
- Supplementaries **accumulate** — none cancels an earlier one; the full chain is retained.
- The **user / lessor** is responsible for keeping the complete set (Initial + all Supplementaries).

Verification procedure — step by step

Step	Action
1	Classify the CoC using the Stage A signals (type tick · Existing Certificate? · installation-type tick).
2	If Initial — record it as the DB's base certificate and proceed to the document/test checks.
3	If Supplementary — read the Previous COC Number and the date it was issued.
4	Locate that Initial in the site set: match the number and confirm it is the same installation / DB / address.
5	Check the dates — the Supplementary must be dated on or after the Initial.
6	Assign the DB status from the matrix below.
7	Assemble the chain for the DB: list the Initial and every Supplementary so the complete current installation is visible (this is the schedule row).
8	Raise follow-ups: request any missing Initial; query any CoC where the Initial/Supplementary box was not ticked.

Decision matrix — what each combination means

What is on file for the DB	Status	Action
Initial only	OK	Base certificate present. Verify it on its own merits (Stages B–D).
Initial + Supplementary(ies), all referencing that Initial	OK	Complete chain. Verify each certificate; keep them filed together.

Supplementary only — references an Initial that IS in the set	OK	Link the two; confirm same installation & date order.
Supplementary only — references an Initial NOT in the set	FLAG	Request the referenced Initial from the client / contractor.
Supplementary only — references no Initial	FAIL	Base installation is uncertified — obtain an Initial CoC.
Multiple Supplementaries, no Initial	FAIL	As above — the chain has no base; obtain the Initial.
CoC present but Initial/Supplementary box not ticked	REVIEW	Determine the type from the original document before filing.
DB requires a CoC but none on file	MISSING	Obtain a CoC for the DB.

Chains of several Supplementaries. Where a DB has an Initial and multiple Supplementaries, confirm each Supplementary points back to the same Initial (or to a prior certificate in the same chain), that dates run forward in time, and that none has been mislaid — a gap in the chain means part of the installation’s alteration history is unaccounted for.

4 · Stage B — Document & admin checks

A CoC is only valid when it is a complete, signed document accompanied by a test report. These are mandatory — a missing item invalidates the certificate.

Check	What must be present
CoC number	A Certificate of Compliance number is recorded.
Test report attached	A test report in the SANS 10142-1 (8.7) format accompanies the certificate.
DB / supply identified	The report states which DB / supply it covers (one report per DB & supply).
Date of issue	Present and valid.
Registered person & number	Section 5 names the registered person with their registration number and type (MIE / IE / single-phase tester).
Signature	Section 5 is signed.
Registration scope	The registration type is adequate for the installation (e.g. a single-phase tester may not sign a three-phase installation).

5 · Stage C — Inspection items (SANS 8.5.2)

The test report’s inspection table has four statements. Per SANS 10142-1 §8.5.2, a **“No” to any of these prevents the certificate from being issued** — so any “No” is a FAIL.

#	Inspection statement	Clause
B1	Conductors are of correct rating & current-carrying capacity for the protective devices and connected load.	8.5.2(a)
B2	Components correctly selected and installed, and accessible where necessary.	8.5.2(b)
B3	Disconnecting devices correctly located; all switchgear switches the phase conductors.	8.5.2(c)
B4	Circuits, fuses, switches, terminals, earth-leakage units, CBs and DBs correctly & permanently labelled.	8.5.2(d)

6 · Stage D — Test values & limits (SANS 8.6)

Each recorded reading is checked against its SANS 10142-1 limit. A reading outside the limit is a FAIL; an illegible or blank reading is “cannot verify” (REVIEW) — never a guess.

#	Test	Clause	Pass limit / criterion
C1	Continuity of bonding	8.6.2	Resistance $\leq 0,2 \Omega$
C2	Earth continuity conductor resistance	8.6.3 / Table 8.1	$\leq$ Table 8.1 value for the protective-device rating (see below)
C3	Continuity of ring circuits	8.6.4	Ring proven continuous (only if ring circuits present)
C4	Earth loop impedance at main switch	8.6.5.1	Low enough to disconnect at $\geq 2 \times$ rated current of the main protective device
C5	Neutral loop impedance	8.6.5.2/.3	Achieves disconnection; never higher than the earth loop impedance (where a neutral is provided)
C6	Prospective short-circuit current (PSCC)	—	Recorded; must not exceed the withstand rating of the main switch & devices
C7	Elevated neutral voltage	8.6.6	$\leq 25 \text{ V}$ pass · $> 25 \text{ V}$ notify supplier · $> 50 \text{ V}$ fail (disconnect)
C8	Insulation resistance	8.6.8.2	$\geq 1,0 \text{ M}\Omega$ (each section/sub-DB also $\geq 1,0 \text{ M}\Omega$). A reading of “infinity” / ∞ / OL / $>999 \text{ M}\Omega$ is the best possible result and passes .
C9	Voltage at DB — no load	8.6.9 / 5.4.2	Within limits: 230 V $\pm 10\%$ (207–253 V) ; 400 V $\pm 10\%$; 525 V $\pm 5\%$
C10	Voltage at DB — on load	8.6.10 / 5.4.2	Within the same 5.4.2 limits at calculated full load
C11	Earth-leakage operation	8.6.11	Sustains at 50% & trips at 100% of $I_{\Delta n}$; socket / personnel circuits $I_{\Delta n} \leq 30 \text{ mA}$
C12	Earth-leakage test button	8.6.12	Unit trips when pressed (each unit)

C13	Polarity at points of consumption	8.6.13	Single-pole devices in phase; correct phase terminals; ES lampholder centre = phase
C14	Phase rotation (three-phase)	8.6.13(d)	Rotation / identification maintained
C15	Switching devices	8.6.14	Circuit interrupted as intended when operated

Table 8.1 — Maximum earth continuity conductor resistance (for C2)

Protective device (A)	6	10	16	20	25	32	40	45	50	63
Max ECC resistance (Ω)	1,533	0,920	0,575	0,460	0,368	0,288	0,230	0,204	0,184	0,146

Final circuits rated > 63 A: use Table 6.28 of SANS 10142-1, not Table 8.1.

7 · How the final verdict is decided

The checks are applied in this order. The first condition that is met sets the verdict:

Order	If...	Verdict
1	No test report is attached / captured	REVIEW confirm attachment
2	Any inspection item (B1–B4) = “No”	FAIL inspection item failed
3	Any legibly-read test value breaches its limit	FAIL value out of limit
4	The test table is blank or unreadable	REVIEW manual check
5	Some key values are illegible (but no breach)	REVIEW values illegible
6	All applicable checks pass	PASS

Golden rule — never fail on what cannot be read. A value that is blank or illegible on the scan is marked “cannot verify” and the CoC goes to REVIEW. A CoC is only failed on a breach that is actually visible (an inspection “No”, or a reading that legibly falls outside its limit). This keeps the audit defensible.

Supplier-notify items are not installation fails. Under SANS 10142-1 the action for an out-of-limit **supply voltage** (C9/C10, clause 5.4.2) or an **elevated neutral voltage > 25 V** (C7, clause 8.6.6) is to *notify the supplier* — these are supply-side conditions, not defects in the installation. They are therefore flagged **REVIEW**, not **FAIL**. (An elevated neutral > 50 V is a disconnect condition and is escalated for urgent action.)

Insulation resistance of “infinity” / ∞ / OL passes. An infinite insulation reading is the best possible result (well above the 1,0 MΩ minimum) and is marked PASS, not cannot-verify.

Only a CORE value forces REVIEW. A COC goes to REVIEW only when a **core mandatory** item cannot be read — signature, test-report present, the four inspection items, bonding (C1), earth-continuity (C2), insulation (C8), earth-leakage trip (C11), test-button (C12), polarity (C13), switching (C15). If only **optional / supplier-notify** fields are blank (earth-loop C4, neutral-loop C5, PSCC C6, elevated-neutral C7, voltages C9/C10, ring C3, phase-rotation C14) the COC still **PASSES** with those fields simply noted. **Bonding/ECC:** a readable low reading passes (a bond is present); only a grossly high reading is failed — exact limit-vs-rating checks are left to the engineer.

8 · The DB schedule status logic

Separately from the per-CoC verdict, each DB on the schedule is given a status based on which certificates exist for it:

- Has an Initial CoC → **OK**
- Only Supplementary CoC(s), none referencing an Initial → **FAIL** (installation has no base certificate)
- Supplementary references an Initial not in the set → **FLAG** (request the Initial)
- CoC required but none on file → **MISSING**
- CoC present but type not ticked → **REVIEW**
- Vacant / not required → **N/A**

9 · File naming convention

Every electrical CoC is renamed to a consistent pattern so the register is self-describing:

<COC number>_<SHOP-TRADING>_<I | S | U>

Examples: ECA371296_SHOP-19B-PEP-CELL_S.pdf · ECA-642715_SHOP-03-POWER-FASHION_I.pdf .
The marker is **I** = Initial, **S** = Supplementary, **U** = type not ticked on the form. Non-electrical certificates (plumbing, glazing, fire, gas, HVAC) are left unrenamed and excluded from the electrical count.

10 · How to read the output files

File (per site)	What it gives you
... DB COC Schedule.xlsx folder 04	One row per DB/tenant with the Initial, Supplementary and Unclear CoC numbers, the supplement→initial reference, file count, status and notes. The register view.

... COC Verification.xlsx folder 04	One row per CoC with a colour grid of every rule (PASS / FAIL / CV / N/A) and an overall verdict. The technical view — “CV” cells are exactly what needs a manual read.
per-COC HTML reports folder 05	A one-page report for each CoC (FAIL / REVIEW / PASS), listing the rules not met, the reason, what could not be verified, and the required action. Open in a browser and Print / Save PDF.
... Verification Findings.html folder 05	A site-level summary: counts and the lists of FAIL and REVIEW certificates.

11 · Limitations & when to check manually

Handwriting. Many test reports are handwritten on scanned forms. Numeric readings frequently cannot be read to audit grade, so the matching CoC is marked REVIEW rather than passed or failed. A person must read those readings off the original certificate and confirm them against the limits in Section 6 of this guideline.

Mixed documents. CoC folders often contain non-electrical certificates (plumbing, glazing/SAGGA, HVAC, fire, gas) and standalone test-report pages. These are filtered out of the electrical count — check the “non-CoC docs” note on the schedule when a DB looks short.

What is reliable without a manual check. The register layer — which CoCs exist, their Initial/Supplementary classification, missing certificates, supplementary-without-initial gaps, and the renamed filenames — is dependable and can be acted on directly. The clear FAILs (an inspection “No”, a visible breach, an unsigned or report-less CoC) are also dependable.

Edition note: this rule set follows SANS 10142-1:2021 (Ed 3.1). A CoC is, strictly, judged against the edition in force on its date of issue; for period-correct verification of older certificates the rule set must be re-pointed to that edition’s clauses.